SDS 8102: Advanced Mathematical Statistics

Fall 2026

Instructor: Qiang Sun (Qiang.Sun@mbzuai.ac.ae) and Saptarshi Roy (Saptarshi.Roy@mbzuai.ac.ae)

Teaching Assistant: Ziyun Zou (Ziyun.Zou@mbzuai.ac.ae)

Lectures: Tuesday and Thursday, 1 pm - 2:30 pm, 1A-B201

Labs: Friday, 2 pm - 3:50 pm, 1B-B201

Office Hours:

Qiang Sun: TBD

Saptarshi: Wed 2pm - 3:30pm, Location: Building 1B, C-1.06.

Zinyu: TBD

Overview

This course explores statistics based on the principles of probability theory. It takes an in-depth look at the mathematical foundations of decision theory and provides students with tools and techniques for constructing estimators, hypothesis tests, and confidence regions. The course also focuses on the exploration of empirical process theory, emphasizing key optimality results of likelihood methods. In addition, the course highlights several key areas of statistics, such as resampling techniques and nonparametric statistics.

The course consists of four sections. In the first section, we review the construction of statistical experiments and the methods for constructing point estimators, hypothesis tests, and confidence intervals. In the second, we provide a concise presentation of statistical decision theory, risks, losses, and discuss the optimality of statistical decisions. In the third section, we present asymptotic statistics, providing elements for constructing the empirical process (uniform laws of large numbers). The fourth section presents to current research areas in statistics, including resampling methods and bootstrap and non-parametric statistics.

Knowledge requisites

- Calculus, real analysis, and linear algebra.
- Measure-based probability theory
- An introductory course on statistics is recommended.

Evaluation

Evaluation will be based on four homeworks, class & lab attendance, one midterm, and one final exam. The grading breakdown is as follows (each homework is worth an equal amount):

Assignment: 15%

Midterm exam: 35%

Final exam: 40%

Class & lab attendance: 10%

Homework

The homeworks are structured to give you experience in primarily written mathematical exercises, and to a lesser extent, computational exercises for additional intuition. Detailed submission instructions will be provided in the homework.

In general, late homework will not be accepted. In the case you are busy preparing for an important deadline, you must give us at least -3 days of notice if you are requesting an extension. In the case of an emergency, no notice is needed, and we can work with you to give you a reasonable extension. But please use this option only in the case of true emergencies.

We encourage discussion and collaboration on homework assignments with other students. However, such collaboration must be clearly acknowledged by listing the names of students with whom you collaborated. We will assume that you are taking full responsibility in terms of writing your own solutions and implementing your own code.

AI use: AI-generated solutions are strictly discouraged. Assignments are designed to prepare you for the exams and it should in your best interest to simulate a similar environment. Please also consult the Academic Integrity Policy.

Take care of your self: All of us benefit from support during times of struggle. You are not alone. Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressures and challenges associated with relationships, mental health, identities, finances, etc. If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. Please also visit the Health, Wellness, and Safety website if needed.